

Targeted Therapy

If your treatment plan includes targeted therapy, knowing how it works and what to expect can help you prepare for treatment and make informed decisions about your care.

What is targeted therapy?

What is the goal of targeted therapy?

How is targeted therapy different from chemotherapy?

Which cancers can be treated with targeted therapy?

What is it like to get targeted therapy?

Targeted therapy safety

What are the side effects of targeted therapy?

What is targeted therapy?

Targeted therapy is a type of **precision** or **personalized medicine** that treats cancer by targeting specific features, changes, mutations, or substances in or on cancer cells.

There are many kinds of targeted therapies. They are designed to stop cancer cells from growing and spreading while limiting damage to normal, healthy cells. Each type works in a specific way. For example, they might:

- Target specific biomarkers (genes and proteins that help cancer cells survive and grow)
- Change the tissue or environment that cancer cells grow in
- Target other types of cells that help a cancer grow, like blood vessel cells

What is the goal of targeted therapy?

If your doctor has recommended targeted therapy, it's important to understand the goal of treatment. This can depend on your type of cancer, its stage (size and location), what biomarkers (gene and protein changes) it has, and how far it has spread.

Targeted therapy can have 3 possible treatment goals:

To cure the cancer (curative intent)

Whenever possible, the goal of targeted therapy is to cure the cancer—destroying it completely so it doesn't come back. Doctors may call this "treatment with curative intent." They may avoid saying "cure" because it can take many years to know for sure if the cancer is truly gone. Cure is the hope, but it's not always possible.

To control the cancer

If a cure is not possible, targeted therapy may be used to slow the growth of the cancer or shrink it. This can help people feel better and live longer. The [cancer might not go away completely¹](#), but it can often be controlled and managed as a chronic disease, like heart disease or diabetes. Sometimes the cancer may go away for a while but come back later.

To ease symptoms (palliation or palliative intent)

Targeted therapy can also be used to help people feel better. This is called **palliation**. It may be given even when the cancer has spread or can't be controlled to improve comfort and quality of life. For example, it may help shrink a tumor that's pressing on an area and causing pain.

Palliative targeted therapy is just one part of [palliative care²](#), which includes things like pain relief and anti-nausea medicine. Palliative treatments can be used at any stage of cancer.

How is targeted therapy different from chemotherapy?

[Chemotherapy](#)³ and targeted therapy are both treatments for cancer, but they are different types of drugs. They work in different ways and cause different side effects.

- Targeted therapy acts on specific types of cancer cells and mostly leaves normal, healthy cells alone. Traditional chemo, however, is **cytotoxic** to most cells, meaning it can damage normal, healthy cells too.
- Some targeted therapies work by blocking cancer cells from copying themselves. This means they can help stop a cancer cell from dividing and making new cancer cells. Traditional chemo, however, kills cancer cells that have already been made.

Both chemo and targeted therapy are **systemic** treatments. They travel through the bloodstream to reach all parts of your body. This means they can both treat cancer that has spread (metastasized) to parts of the body far from where the original (primary) tumor started, or blood cancers, such as leukemia, that have spread throughout the body. This makes them different from **local** treatments like surgery and radiation, which only affect one part of the body.

How Does Targeted Therapy Work?

Some targeted therapies act on specific aspects that make cancer cells unique. Others block certain messages sent inside a cancer cell that tell it to grow. Learn more.

Which cancers can be treated with targeted therapy?

Targeted therapies are not yet available for all cancers. The biomarkers (gene changes, proteins, and other substances) these treatments target can be different for different cancers, and even for different people with the same type of cancer. Targeted therapy is most often used for:

- Breast cancer
- Colorectal cancer
- Gastrointestinal stromal tumors
- Leukemias (acute, chronic, or specific subtypes)
- Liver (hepatocellular) cancer
- Lung cancer
- Melanoma
- Ovarian cancer

- Pancreatic cancer
- Prostate cancer
- Stomach cancer
- Thyroid cancer

[Biomarker testing](#)⁴ is often needed to know whether your cancer might respond to targeted therapy.

Learn about targeted therapy for your cancer type

Choose a cancer type Acute Lymphocytic Leukemia (ALL) Acute Myeloid Leukemia (AML) Basal and Squamous Cell Skin Cancer Bile Duct Cancer Bladder Cancer Bone Cancer Brain and Spinal Cord Tumors Brain and Spinal Cord Tumors in Children Breast Cancer Cancer of Unknown Primary Cervical Cancer Chronic Lymphocytic Leukemia (CLL) Chronic Myeloid Leukemia (CML) Colon and Rectal Cancer Endometrial Cancer Esophageal Cancer Gallbladder Cancer Gastrointestinal Stromal Tumor (GIST) Kidney Cancer Laryngeal and Hypopharyngeal Cancer Leukemia in Children Liver Cancer Melanoma Mesothelioma Nasal Cavity and Paranasal Sinus Cancer Nasopharyngeal Cancer Neuroblastoma Non-Hodgkin Lymphoma Non-small Cell Lung Cancer Oral Cavity and Oropharyngeal Cancer Ovarian Cancer Pancreatic Cancer Pancreatic Neuroendocrine Tumor (NET) Prostate Cancer Salivary Gland Cancer Soft Tissue Sarcoma Stomach Cancer Thyroid Cancer Uterine Sarcoma Vaginal Cancer Waldenstrom Macroglobulinemia

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What is it like to get targeted therapy?

Most people get targeted therapy as infusions or injections at a clinic or hospital or by taking medicine by mouth at home. Each type has its own routine, side effects, and safety steps.

Targeted therapy schedules

Some targeted therapy is given at regular intervals called **cycles**. A cycle may be a dose of targeted therapy (sometimes with other treatments) on one or more days, followed by several days or weeks without treatment. You may also take targeted therapy every day with no rest breaks, or only 1 or more days a week or month.

Each targeted therapy is given on a schedule that makes the most of its anti-cancer actions and minimizes side effects. If you are getting more than one treatment, your treatment plan will say how often and exactly when each should be given.

Intravenous (IV) targeted therapy

Many types of targeted therapy are given as an infusion through an IV. The process is much like [getting chemo infusions](#)⁵. If you are getting a combination of chemo and targeted therapy, they might be given at the same time or during separate appointments, depending on your treatment plan.

Subcutaneous (SC or SQ) targeted therapy

Subcutaneous (SC or SQ) targeted therapy is given as an injection or shot into the **subcutaneous tissue** (fatty tissue) right under your skin (similar to [subcutaneous chemo](#)⁶). Not all types of targeted therapy can be given as a SQ injection, but the number is increasing. Giving targeted therapy this way is just as effective, but takes less time and doesn't require an IV.

Oral targeted therapy

Oral targeted therapy is targeted therapy that can be taken by mouth as pills, capsules, or liquids. Most **small-molecule** targeted therapies are oral. Examples include imatinib, sunitinib, lenalidomide, and everolimus. It may be given in rounds or cycles and is often given along with traditional chemotherapy (chemo). Because oral targeted therapy is taken at home, it's important to understand exactly how and when to take it, and what to do if you have any problems. Our information on [oral chemotherapy](#)⁷ can help you know what to expect and what to ask your care team.

Targeted therapy safety

Many targeted therapies, including small-molecule drugs and a few monoclonal antibodies, are considered hazardous agents that require special handling, similar to chemotherapy. But because many targeted therapies are newer, there isn't as much information about the long-term effects of exposure. It's important to follow the safety precautions recommended by your cancer care team. Usually, these are the same [precautions used for chemotherapy safety](#)⁸.

What are the side effects of targeted therapy?

Targeted therapy has different side effects than standard or traditional chemotherapy (chemo). Some targeted therapies have very few side effects, and others can cause more or more serious problems. There are many different types of targeted therapies, and their side effects depend largely on the type that's given and what it targets.

Targeted Therapy Side Effects

There are many different types of targeted drugs to treat cancer. Read here about their side effects and how long the effects might last.

Hyperlinks

1. www.cancer.org/cancer/survivorship/long-term-health-concerns/cancer-as-a-chronic-illness.html
2. www.cancer.org/cancer/supportive-care/palliative-care.html
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8. www.cancer.org/cancer/managing-cancer/treatment-types/chemotherapy/chemotherapy-safety.html

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How Does Targeted Therapy Work?

There are different types of targeted therapy, and each works in its own way. For certain cancers, testing can be done to see if any targeted therapies might be available. If so, targeted therapy might be used on its own or along with other types of treatment.

[How do targeted therapies target cancer cells?](#)

Which biomarkers can be treated with targeted therapy?

Examples of targeted therapies

Choosing the right targeted therapy

How does targeted therapy work with other treatments?

How successful is targeted therapy?

When treatment plans change

How do targeted therapies target cancer cells?

Targeted therapies are made to find and attack specific areas or substances in or on cancer cells. Or they can detect and block certain kinds of messages sent inside a cancer cell that tell it to grow. A targeted therapy might target:

- Too much of a certain protein on a cancer cell
- A protein on a cancer cell that is not on normal cells
- A protein that is mutated (changed) in some way on a cancer cell
- Gene (DNA) changes that aren't in a normal cell

These gene and protein changes are called [biomarkers](#)¹.

Each targeted therapy has a specific action it takes on cancer cells. Depending on the biomarker it targets, it can:

- **Block or turn off chemical signals** that tell the cancer cell to grow and divide
- **Change proteins** within the cancer cells so the cells die
- **Stop the formation of new blood vessels** that feed the cancer cells
- **Trigger your immune system** to kill the cancer cells
- **Carry toxins to the cancer cells** to kill them, but not normal cells

The action a targeted therapy has on cancer cells affects where these medicines work best and what side effects they may cause. Targeted therapies are broadly classified into 2 groups based on the way they work: **small-molecule drugs** and **monoclonal antibodies**. Some work in more than one way.

What are small-molecule drugs?

Targeted therapies called **small-molecule drugs** are lab-made chemicals that are tiny enough to enter a cancer cell once they find it. They work by targeting a specific biomarker or substance inside the cell and blocking it from working. This helps prevent cancer cells from growing and spreading.

The generic names of small-molecule drugs often end in **-ib**, such as imatinib.

What are monoclonal antibodies?

Monoclonal antibodies are proteins made in a lab. They are too big to enter the cancer cell directly. Instead, they aim for and act on the surface of the cell or in the area around the cell to block a specific target. Some monoclonal antibodies work by themselves. Others carry chemotherapy or other treatments to the cancer cells to help destroy them.

The generic names of monoclonal antibodies often end in **-mab**, such as rituximab.

Most monoclonal antibodies are considered targeted therapy, but some work in a different way, acting on your immune system to help it find and attack cancer cells more effectively. Learn more about [monoclonal antibodies that affect the immune system](#)².

Which biomarkers can be treated with targeted therapy?

All cancers start because of [gene changes](#)³ (mutations) that turn normal cells into cancer cells. But cancers can have many different types of mutations or other biomarkers. Even when two people have the same type of cancer (like colon or breast cancer), their cancers may have different biomarkers that affect which treatments might work best.

Some of the more common gene or protein changes in cancer cells that have targeted therapies approved for treatment include:

- *ALK genes* (some lung cancers or lymphomas)
- *BCR-ABL protein* (some chronic myeloid leukemias)
- *BRAF genes* (some colorectal, lung, thyroid cancers, and melanoma)
- *BRCA* (like *BRCA1* and *BRCA2*) *genes* (some breast, ovarian, pancreatic and prostate cancers)

- CDK4/6 proteins (some breast cancers)
- *C-KIT gene* (some leukemias and gastrointestinal stromal tumors)
- *EGFR genes* (some colorectal and lung cancers)
- *HER2/neu (ERBB2) genes* (some breast, colorectal, lung, ovarian, pancreatic and stomach cancers)
- *KRAS genes* (some colorectal, lung and pancreatic cancers)
- *MET genes* (some lung, thyroid, and renal cell cancers)
- *mTOR protein* (some kidney cancers)
- *NTRK genes* (some colorectal, lung, and pancreatic cancers)
- *PARP protein* (some ovarian cancers)
- *PIK3CA genes* (some breast, colorectal, lung, and ovarian cancers, or lymphomas and chronic leukemias)
- *RET genes* (some lung and thyroid cancers)
- *ROS1 genes* (some lung cancers)
- *VEGF genes* (some colorectal, hepatocellular, lung, renal cell, and thyroid cancers)

Examples of targeted therapies

In addition to being classified as small-molecule drugs or monoclonal antibodies, targeted therapies can also be grouped based on their specific target or the way that they act against the cancer cell. **Some can act in multiple ways and are included in multiple groups.** Different types can cause different side effects depending on the specific way they work. Here are some examples:

Angiogenesis inhibitors (anti-angiogenesis drugs)

How they work: To get the blood and nutrients they need to grow, tumors send signals that cause new blood vessels to form nearby. This process is called **angiogenesis**. Treatment with **angiogenesis inhibitors** helps stop the formation of new blood vessels by targeting proteins such as vascular endothelial growth factor (VEGF). Blocking this protein can slow or stop cancer growth by limiting its blood supply.

Angiogenesis inhibitors are monoclonal antibodies, so their generic names often end with **-mab**, such as bevacizumab.

Examples of angiogenesis inhibitors:

- **Bevacizumab** is used in combination with some types of traditional chemo, immunotherapy, or other targeted therapy. It can be used for some types of cervical, colorectal, liver, non-small cell lung, and ovarian cancers, as well as glioblastoma (a type of brain cancer) and renal cell carcinoma (a type of kidney cancer). Side effects include bleeding, headache, high blood pressure, kidney problems, delayed wound healing, and fatigue.
- **Ramucirumab** is used in combination with some other targeted therapy or traditional chemo for management of colorectal, liver, non-small cell lung, or stomach cancer. Side effects include bleeding, headache, high blood pressure, kidney problems, delayed wound healing, and fatigue.

Kinase inhibitors

How they work: Kinase inhibitors block the action of **kinase**, a type of protein inside cells that normally helps tumor cells grow. Some of the known kinases that respond to kinase inhibitors include KRAS, EGFR, ALK, PIK3, and BRAF. Blocking these kinases can often stop the growth of the cancer. There are different groups of **kinase inhibitors**, based on the kinds of proteins they block: tyrosine kinase inhibitors (TKIs), serine/threonine kinase inhibitors, and nonspecific inhibitors.

Kinase inhibitors are small-molecule drugs. Their generic names often end in **-nib**, such as imatinib.

Examples of kinase inhibitors:

- **Osimertinib** is a tyrosine kinase inhibitor that targets cancer cells with too much of a protein called epidermal growth factor receptor (EGFR). It blocks EGFR to keep the cancer cells from growing faster. Osimertinib can be used in cancers like non-small cell lung cancer that have certain abnormal *EGFR* genes. Side effects include low white blood cell count (neutropenia), low platelet count (thrombocytopenia), skin changes (rash), diarrhea, nail changes, cough, and lung problems.
- **Imatinib** is another type of tyrosine kinase inhibitor that targets cancer cells with the BCR-ABL protein. This protein sends out signals for cells to grow and divide without stopping. Blocking it slows cancer growth. Imatinib can be used for some types of chronic myeloid leukemia or gastrointestinal stromal tumors. Side effects include fatigue, edema, nausea, diarrhea, and muscle cramps.

Proteasome inhibitors

How they work: Proteasome inhibitors work inside cells by blocking enzyme complexes called **proteasomes** from breaking down the proteins that keep cell division under control. This disrupts normal cell functions, especially in cancer cells, which produce more abnormal proteins and are more likely to die as a result.

Proteasome inhibitors are small-molecule drugs, with generic names often ending in **-mib**, such as bortezomib.

Examples of proteasome inhibitors:

- **Bortezomib** is used to treat multiple myeloma, often in combination with other targeted therapy and steroids. It can also be used to treat some lymphomas, usually after other treatments have been tried. Side effects include skin changes, diarrhea, nausea, low blood counts and nerve damage.
- **Carfilzomib** is also used to treat multiple myeloma. Like bortezomib, it is used in combination with other targeted therapies. It can also be used to treat [Waldenstrom macroglobulinemia](#)⁴ (WM). Side effects include high blood pressure, edema, diarrhea, nausea, low red blood counts, fatigue, and headache.

Signal transduction inhibitors

How they work: Signal transduction is the way cells act on signals from hormones and growth factors outside the cell. This process helps normal cells function properly. But cancer cells can alter the signal pathways, allowing them to grow and divide more easily or avoid dying. **Signal transduction inhibitors** disrupt these abnormal cell signals to help stop the cancer from growing.

Signal transduction inhibitors are small-molecule drugs. Their generic names can have different endings, including **-sib** and **-nib**.

Examples of signal transduction inhibitors:

- **Idelalisib** is a PI3K inhibitor. It targets a protein called phosphatidylinositol 3-kinase that sends signals affecting cell growth. It is used to help treat chronic lymphocytic leukemia after other treatments have been tried. Side effects include skin changes, fever, fatigue, diarrhea, cough, low blood counts, and nausea.

- **Dabrafenib** is a BRAF inhibitor that acts on changes in the BRAF protein. About half of all melanomas have changes in the BRAF gene, allowing them to make an altered protein that helps them grow. Dabrafenib is often used along with other targeted therapies to treat melanoma that has spread or can't be removed completely. Side effects include skin changes, fever, fatigue, and nausea.

Other targeted therapies

There are many other types of targeted therapies that work in different ways. Some examples include:

- Antibody-drug conjugates
- Apoptosis inducers
- Gene expression modulators
- Hedgehog inhibitors
- Hormone therapies
- Immunotherapies
- mTOR inhibitors
- PARP inhibitors
- Toxin delivery molecules

Learn about targeted therapy for your cancer type

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Choosing the right targeted therapy

Choosing the right targeted therapy is very important. Targeted therapy is sometimes called **precision** or **personalized medicine** because even people with the same type of cancer can have different biomarkers. Different targeted therapies are used depending on which biomarkers are found through testing.

Biomarker testing

For certain types of cancers, [biomarker testing](#)⁵ is done (usually after a biopsy or surgery) to look for biomarkers that could be targeted.

[Watch this video on YouTube](#) ⁶

Not all known biomarkers have effective treatments yet, and many are still being studied.

- If a targetable biomarker is found, doctors can match the person with a treatment that is more likely to work.
- If a cancer doesn't have the biomarker that a targeted therapy is designed to act on, that treatment won't be effective.

Treatment options

Factors a cancer care team considers when recommending treatment options include:

- The type and subtype of cancer
- The stage of the cancer (where it is, how big it is, and whether it's spread to other parts of the body)
- Results of biomarker testing
- Your age
- Your overall health and current medications
- Other serious health problems (such as heart, liver, or kidney diseases)
- Goal of treatment
- Types of cancer treatments you have had in the past

The cancer care team also considers the standard-of-care treatment guidelines for specific cancers. They may also offer a [clinical trial](#)⁷ if one is available and the person is eligible for the trial.

If your treatment plan is not clear, or you are given several possible options, ask your cancer care team about [getting a second opinion](#)⁸. It's important to feel confident that you understand your choices and are making the decision that's right for you.

How does targeted therapy work with other treatments?

So far, only a few types of cancer are routinely treated using only targeted therapy or a combination of targeted therapies.

More often, targeted therapy is used along with other treatments. Some targeted therapies are even specifically designed to help other treatment work better. For example, they might make cancer cells more sensitive to chemo or block the cancer's ability to repair damage caused by chemo.

Targeted therapy may be used:

- To shrink a tumor before surgery or radiation therapy **neoadjuvant therapy**)
- After surgery or radiation therapy to help kill any remaining cancer cells in the body (**adjuvant therapy**)
- With other types of medicines, such as [traditional chemo](#)⁹, [immunotherapy](#)¹⁰, or [hormone therapy](#)¹¹
- With other treatments if cancer comes back or doesn't completely go away

How successful is targeted therapy?

Targeted therapy can be very successful at treating cancer. However, there are times it may not work or may stop working. This is usually because:

- **The cancer doesn't have the specific biomarker** for the targeted therapy to find and work against. The exact target needs to be present in the cancer cells for a targeted therapy to work. This is why targeted therapies can't be used for everyone with a certain type of cancer.

- **The cancer develops drug resistance.** This happens if the cancer cells develop new changes or mutations. If the new change or mutation is not a target for the medicine, the drug will no longer be effective. This is a common reason for targeted therapies to stop working and in many cases, can be tested for by getting another biopsy sample.
- **People can respond differently to a targeted therapy.** Any variation from person to person affects how well the targeted therapy works against their cancer.

Each person's cancer is unique, and differences in their genes, type and makeup of cancer, overall health, and how they respond to treatment can all impact the success of targeted therapy. Doctors and scientists are still learning about these drugs and personalized medicine remains a very personal experience.

When treatment plans change

Doctors use guidelines based on research and clinical trials to know the best treatment options. It's important to get the full prescribed course of targeted therapy to get the most benefit from treatment. But this isn't always possible.

Sometimes, the treatment plan must be changed. You might develop drug resistance, meaning the targeted therapy stops working. If this happens, your doctor will talk with you about other treatment options.

You may also develop another health problem or serious side effects that require adjusting the dose or schedule to help your body recover. **Supportive medicines** may also be used to help manage side effects. The goal is to give enough targeted therapy to treat the cancer without causing other serious problems.

In some situations, targeted therapy may be recommended after other types of treatment has been unsuccessful in slowing the growth of the cancer. Learn more about options [if cancer treatments stop working](#)¹² or whether a [clinical trial](#)¹³ may be an option for you.

Hyperlinks

1. www.cancer.org/cancer/diagnosis-staging/tests/biomarker-tests.html

2. www.cancer.org/cancer/managing-cancer/treatment-types/immunotherapy/monoclonal-antibodies.html
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Targeted Therapy Side Effects

Targeted therapy has different side effects than standard or traditional chemotherapy (chemo). Some targeted therapies have very few side effects, and others can cause more or more serious problems. There are many different types of targeted therapies, and their side effects depend largely on the type that's given and what it targets.

[Why does targeted therapy cause side effects?](#)

[How bad are the side effects from targeted therapy?](#)

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Why does targeted therapy cause side effects?

Targeted therapy works by acting on specific biomarkers such as genes or proteins that are mainly found in cancer cells. This limits damage to other normal, healthy cells. But because healthy cells can also have some of these proteins, targeted therapy can affect them too, causing side effects.

The normal cells most likely to be damaged by targeted therapy are:

- Skin and nail cells
- Hair follicles
- Cells in the mouth, digestive tract, and reproductive system
- Blood-forming cells in the bone marrow

Some targeted therapies can also damage cells in the heart, lungs, kidneys, and thyroid.

How bad are the side effects from targeted therapy?

Side effects are not always as bad as you might expect, but it's normal to worry about this part of cancer treatment.

Sometimes, you can take medicines with the targeted therapy to help protect your body's normal cells. There are also treatments to help relieve side effects.

Doctors try to give targeted therapy at levels high enough to treat cancer, while keeping side effects at a minimum. They also try to avoid using multiple treatments that have similar side effects.

Things to remember:

- Not every person gets every side effect, and some people get few or none.
- How severe a side effect might be can vary greatly from medicine to medicine and from person to person.
- Rare and unusual side effects can happen with some targeted therapies, and some can be serious. Report all changes and side effects as soon as possible to your cancer care team.

Be sure to talk to your cancer care team about:

- Which side effects are most common with your treatment
- How long they might last
- How bad they might be
- When you should call the doctor's office about them.

Your doctor may give you instructions to follow or medicines to help prevent some side effects before they happen.

How long do targeted therapy side effects last?

Most side effects go away over time once treatment ends and your normal cells recover. The time it takes to get over some side effects varies from person to person. It depends on many factors, including your overall health and the targeted therapy you were given.

Because many targeted therapies are still quite new, it's hard to say how long you can expect side effects to last. We do know that some of the side effects from traditional chemo can last a lifetime, such as when it causes long-term damage to the heart, lungs, kidneys, or reproductive organs. In many cases we still don't know if targeted therapy causes these kinds of long-term changes.

What are common side effects of targeted therapy?

Side effects of targeted therapy can be very different depending on the type of targeted therapy you are getting. Some occur more often than others, and some can be quite serious. They also are treated differently than side effects from other types of cancer treatment. Talk with your care team about the possible side effects of the targeted therapy you will be taking, what to look for, and how to take care of yourself if they occur.

Some of the other common and serious side effects caused by targeted therapy are listed here. This is not a complete list, as each targeted therapy can have different side effects.

Skin, hair, and nail changes

Many targeted therapies cause skin, hair, and nail changes (including hair loss). These problems usually develop slowly over days to weeks after your treatment starts. They **are not** signs of a drug allergy.

Why do skin changes occur?

[Skin changes](#)¹ are caused by the way some targeted therapy works. For example:

EGFR inhibitors such as cetuximab or erlotinib attack the EGFR (epidermal growth factor receptor) protein, which tells the cancer cells to grow and divide. Skin cells also have a lot of EGFR, so these therapies affect them too. They turn off the signal for skin cells to grow normally and make it harder for them to retain moisture.

Angiogenesis inhibitors like bevacizumab target VEGF (vascular endothelial growth factor) proteins. Blocking these proteins helps stop tumors from building and keeping a blood supply, but it also affects the very small blood vessels in the hands and feet. This can cause hand-foot syndrome (described later).

What is the most common skin change?

Rash is the most common skin change from targeted therapies and depends on the type and dose of the targeted therapy. In most people, the rash is mild. It often looks like acne and shows up on the scalp, face, neck, chest, and upper back. In severe cases, it can affect other parts of the body.

The rash most often starts as skin redness and swelling. It's often worst within the first few weeks of treatment. By about a month into treatment, the skin usually crusts and gets very dry and red. In the weeks after that, round, flat or raised red spots and pimples with pus in the center often appear. In some people this can lead to skin infections. The rash can itch, burn, or sting and may be painful. It may get better on its own or stay about the same during the rest of treatment, but it should go away completely about a month after treatment is stopped.

The rash can be very distressing and make a person feel self-conscious around others. There are some medications that your doctor can try to make you feel better while you have the rash. Prevention and treatment of rashes are discussed later.

What other kinds of skin changes should I watch for?

- **Changes in how your skin feels:** Your skin may start to feel like it's sunburned, before any redness or rash shows up. This can occur as early as the first week of treatment.

- **Photosensitivity:** Your skin will likely become much more sensitive to light and more easily damaged by UV rays during treatment. This makes it easier for your skin to burn and blister, even after very little sun exposure or exposure to sun coming through windows.
- **Dry skin:** This is very common for many targeted therapies. It can start within the first few weeks, but nearly everyone getting targeted therapy has [dry skin](#)² after 6 months of treatment. The skin can become very dry, brittle, itchy, and scaly and may even crack open—especially on the hands and feet. Cracking can happen by itself, but it tends to be worse when there's also a rash.
- **Itching:** Many skin changes, like rash or dryness, can cause [itching](#)³.

Hand-foot syndrome (HFS): [HFS](#)⁴ usually starts during the first 2 to 6 weeks of treatment. Painful sensitivity, tingling, or numbness in the hands and feet are the earliest symptoms. Then, the palms of the hands and the soles of the feet become red and swollen. The redness looks a lot like sunburn and may blister. In severe cases, the blisters can open up and become sores. The affected skin also can become dry, peel, and crack.

What kinds of nail and hair changes should I watch for?

- **Changes in the nails:** Some targeted therapies can cause [swollen, red, and painful sores](#)⁵ around the fingernails and toenails. (This can look a lot like an infection or an ingrown nail.) This most often happens to the big toes and thumbs. These sores may become infected. Nails may also become brittle and grow more slowly.
- **Sores in areas of hair growth:** Some people notice sores on their scalp and on other hairy areas. Scars caused by these sores may keep your hair from growing back after treatment.
- **Changes in hair growth:** Some targeted therapies can cause the hair on your head to become thin, dry and brittle, or even curly. Long-term use may lead to [bald patches or complete loss](#)⁶ of scalp hair. Facial hair may grow faster than usual, including longer, thicker, curly eyebrows and eyelashes that may need to be trimmed. But in some men, facial hair growth slows down. Eyebrows may thin out as well.
- **Changes in hair or skin color:** Some targeted therapies can turn the [skin or hair a yellowish color](#)⁷ during treatment. In a few people, hair and/or skin gets darker. This tends to go away once treatment ends.

- **Changes in and around the eyes:** The eyes may burn and become dry or red. In some people, the eyelids get red, tender, and swollen, and the lashes may become crusty. Sometimes the eyelids may turn inward or outward. Distorted eyelids or prolonged dryness can damage the outer part of the eye (the cornea). Talk with your doctor or nurse about managing these changes to avoid injury, pain, or infection.

Can skin, nail, and hair changes be prevented?

There are things you can do to help prevent some of these changes or at least try to keep them under control. Your doctor may ask you to start doing these things as soon as targeted treatment starts, before you have problems.

Having good skin, nail and hair care habits before side effects happen may help to minimize the problems. You may be asked to:

- Use mild soaps, body washes, and shampoos that do not contain alcohol, perfume, or dye.
- Take baths instead of showers, and try oatmeal bath products to soothe your skin.
- Bathe with cool or lukewarm (instead of hot) water, and avoid hot, humid places.
- Moisturize your skin at least twice a day with a thick emollient cream that has no alcohol, perfumes, or dyes. The best time to do this is right after you bathe, while your skin is still damp.
- Wear loose, soft clothing.
- Keep nails short.
- Use laundry detergents and fabric softeners without strong perfumes.
- Stay out of the sun as much as possible because sunlight seems to trigger or worsen rashes in some people. If you'll be outside during the day, wear a hat and clothes with long sleeves. Use a broad-spectrum sunscreen with SPF of at least 30 and zinc oxide or titanium dioxide at least 1 hour before going out. Be careful near windows too.
- **Avoid acne medicines.** Though the rash may look like acne, acne medicines don't work. They can even dry it out and make it worse.
- Try gel shoe inserts if the soles of your feet are tender.
- Wear shoes that fit well and aren't too tight. Thick, soft socks may help if your shoes are big enough for the extra bulk.

Ask your doctor or nurse if there are other things you can do to help lower your chance of skin, nail, or hair problems.

What should I do if I have skin, nail, or hair changes?

It's very important to tell your cancer care team right away if you notice any rashes or other skin, nail, or hair changes. They can recommend ways to manage these problems. Left untreated, rashes can get worse and lead to infections, which might then lead to delaying or even stopping treatment.

Don't treat your skin with over-the-counter medicines or stop taking your targeted therapy without talking to your doctor first. Your doctor may give you a skin cream or a medicine to take by mouth to treat the skin.

Be sure to let your doctor or nurse know if:

- You notice a burning feeling, redness, or rash. Depending on the severity and the specific symptoms you are having, there are creams (including steroids or antibiotics) you can use to try to keep it from getting worse.
- Your skin is dry, flaking, or cracked. Moisturizing cream may help with this.
- Your skin is itchy. There are moisturizing or anti-itch creams and gels you can use to ease itching. There are also some medicines you can take by mouth to try to stop the itching.
- The area around your fingernails or toenails becomes sore or red. Moisturizing creams and warm water soaks can help with this. But you and your cancer care team will need to watch for changes that could be signs of infection, which need to be treated quickly.
- You have very dry, red, or tender eyes, or if you notice eyelashes growing inward toward the eyeball.
- You get sores on your scalp or other areas with hair. You'll want to get them treated to help prevent scars that may keep hair from growing later.

Your doctor may tell you to avoid direct exposure to sunlight when possible. Even after treatment is over, you may find that you're more sensitive to sunlight than before. This can last from a few weeks to months after treatment ends.

Can skin changes be treated?

If you have skin changes, your doctor will need to check your skin fairly often to figure out the problem, the best course of action, and whether treatment is helping. You'll probably need extra doctor visits while the problem is being brought under control.

Mild changes: These include rashes that are only in a limited area, that are not causing any distress, and are not infected. These might not need any treatment, but heavy skin creams or ointments that contain no alcohol, perfume, or dye can sometimes help with dryness. Be sure to talk with your cancer care team before using anything on your skin.

Your doctor may prescribe a mild steroid cream or antibiotic gel to put on the rash.

If your eyelids are crusty or swollen, careful cleansing and clean, warm, wet cloths laid over your closed eyes may help.

For mild skin problems, the dose of the targeted therapy usually does not need to be changed. You'll be watched closely to see if the rash gets better or worse.

Moderate changes: These include a rash over a larger area of the body or skin changes causing mild distress from itching or soreness, but with no signs of infection. Your skin may be treated with a prescription cream or gel. Your doctor may also prescribe an antibiotic you take by mouth.

For eye problems, drops or ointments may be prescribed.

The dose of the targeted therapy usually does not need to be changed for moderate skin problems. Still, you will be watched closely to see if the rash gets better or worse.

Severe changes: These are bad rashes that cover a lot of skin, cause itching and soreness that affect your quality of life (such as sleep problems or pain), and are likely to get infected. Treatment is much like that used for moderate changes, including creams or gels, as well as an antibiotic that's taken by mouth. Along with this, a course of steroid pills is often given.

The targeted therapy dose often needs to be reduced when a person has severe skin changes. Expect to see your doctor often during this time. If the rash doesn't get better within about 2 weeks, the targeted therapy is often stopped until the skin changes improve. It may then be re-started with continued skin care.

A note about steroid skin creams and gels

Steroids that are spread on the skin in the form of creams, ointments, or gels can help many skin problems. But it's important to know that using steroid creams for too long can actually cause other skin problems, and can make you more likely to get a skin infection. For this reason, only use steroid creams (even those that don't require a prescription) as directed by your doctor.

Allergic or infusion reactions

Monoclonal antibodies have a higher risk of causing infusion reactions than other types of targeted therapy.

Allergic or [infusion reactions](#)⁸ tend to start suddenly, usually within minutes to hours after taking the targeted therapy. They may include hives (raised skin welts that often go away in a day or so) and intense itching. An allergic or infusion reaction often includes other serious symptoms such as trouble breathing, dizziness, tightness in the throat or chest, or swelling of the lips or tongue.

If you have these kinds of symptoms, get emergency help and call your doctor right away.

High blood pressure

Some targeted therapy, especially **angiogenesis inhibitors**, can raise your blood pressure. There isn't really anything you can do to prevent this, but your doctor will watch your blood pressure closely if you're getting a medicine that can cause this side effect. Some people need medicine to bring their blood pressure down to safe levels during treatment. They should stay on this medicine until their doctor tells them it can be stopped.

Bleeding or blood clotting problems

Targeted therapies that interfere with new blood vessel growth, such as angiogenesis inhibitors, can lead to problems with bruising and bleeding. These problems are not common and do not happen to everyone. But it can help to be aware of them because there's no known way to prevent them.

Bleeding, such as from the stomach and intestines, can be severe and even life-threatening. Tell your doctor if you throw up blood or material that looks like coffee grounds, or if you notice dark or black stools or bright red blood in your stool. These can be signs of bleeding in the stomach or intestines.

Some targeted therapies can also cause [blood clots](#)⁹ in the lungs and legs, as well as heart attacks and strokes. Let your doctor know if you have problems with sudden swelling, pain, or tenderness in the arm or leg. If you have chest pain, sudden shortness of breath, vision problems, weakness, seizures, or trouble speaking, get emergency help. These can be symptoms of serious problems caused by blood clots.

Slow wound healing

By blocking new blood vessel growth, some of these therapies interfere with wound healing. This can lead to old wounds (cuts) opening up again and new wounds not closing. It can also lead to holes (called perforations) opening up in the stomach or intestine. Tell your doctor right away if you have pain in your belly or vomiting.

Because some of these targeted therapies can affect wound healing, they usually need to be stopped before any planned surgery, including dental procedures. Talk to your cancer doctor as soon as you know about a planned surgery or procedure so you can find out what to do.

Heart damage

Some targeted therapies can damage the heart, especially if used with certain medicines. Your doctor may test your heart function before starting treatment. Possible symptoms of heart damage might include chest pain, increased coughing, trouble breathing (especially at night), rapid weight gain, dizziness, fainting, or swelling in the ankles or legs.

Autoimmune reactions

Certain targeted therapy works by basically taking the brakes off the body's immune system. This can lead to serious side effects if the immune system starts to attack healthy parts of the body. In some people this can cause serious reactions in the lungs, intestines, liver, skin, eyes, nerves, hormone-making glands, or other organs. This isn't common, but for some people, it can be life-threatening.

Swelling

Some targeted therapies cause facial swelling, especially around the eyes. They can also cause [swelling](#)¹⁰ in the feet and legs, as well as the hands. This usually doesn't need to be treated, but a diuretic (water pill) may be used in severe cases.

Other side effects

Other side effects have also been linked to treatment with some targeted therapies. Many of these side effects are the same as those seen with traditional chemo and include:

- [Nausea and vomiting](#)¹¹
- [Diarrhea](#)¹² or [constipation](#)¹³
- [Mouth sores](#)¹⁴
- [Shortness of breath](#)¹⁵ or trouble breathing
- Cough
- Feeling tired all the time ([fatigue](#)¹⁶)
- Headache
- [Hair loss](#)¹⁷
- Damage to organs such as the thyroid gland, liver, or kidneys
- Neutropenia (low [white blood counts](#)¹⁸)
- Increased risks of certain [infections](#)¹⁹
- [Second cancers](#)²⁰

Your cancer care team will watch you closely during treatment and will check you often. Side effects can and should be treated as early as possible. It's important that you tell your cancer care team about any changes in how you feel or anything you notice that's new or unusual. Tell them right away so they can treat any problems and try to keep them from getting worse.

Targeted therapy food and drug interactions and side effects

When planning for targeted therapy, doctors must look at interactions between targeted therapy and other medicines the person is taking, including over-the-counter medicines, foods, vitamins, and supplements. These interactions may make side effects worse or affect how well targeted therapy works. For example:

- Some types of targeted therapy can increase your risk of bleeding. Aspirin and similar medicines can also increase this risk. Taking these together might put some people at risk for a serious bleeding problem.
- Certain foods and drinks, such as grapefruit juice, can interfere with the absorption or effectiveness of some targeted therapies.

Always check with your doctor about the safety of other medicines, vitamins, herbs, and supplements, as well as potential food and drink interactions while you are being treated for cancer. It's important that they know about everything you are taking, even if it's only "as needed."

When to call your cancer care team about targeted therapy side effects

Because your cancer care team will give you lots of information about side effects, you might be more aware of physical changes. Do not take any physical symptoms you have lightly. Some side effects are short-lived and minor, but others may be a sign of serious problems. **Make sure you know how to reach someone on your team any time, including after hours, weekends, and holidays.**

Contact your cancer care team right away if you have any of the following symptoms during targeted therapy treatment:

- A fever higher than what your cancer care team has instructed (usually 100.5°F -101°F or higher, taken by mouth)
- Bleeding or unexplained bruising
- A rash
- An allergic reaction, such as swelling of the mouth or throat, severe itching, trouble swallowing
- Intense chills

- Pain or soreness at the targeted therapy injection site or catheter site
- Unusual pain, including intense headaches
- Shortness of breath or trouble breathing (If you're having trouble breathing, call 911 first.)
- Long-lasting diarrhea or vomiting
- Bloody stool or blood in your urine

Ask your cancer care team if there are any other problems they should know about right away.

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/rashes-skin-changes.html
2. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/rashes-skin-changes.html
3. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/rashes-skin-changes.html
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